A Tale of Water and Fish: Untangling the Capitalization Value of Bundled Ecosystem Services in Freshwater Lakes

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Abstract

Hedonic housing market valuation of freshwater ecosystems often focuses on aesthetic water quality, overlooking the capitalized value of recreational fisheries. This can be problematic in cases where nutrients simultaneously degrade water quality by supporting increased phytoplankton growth and stimulating the productivity of prized game fisheries. We demonstrate that omitting fish populations from standard hedonic models generates omitted variable bias in mesotrophic lake systems, biasing estimates of the economic value of water clarity. Leveraging 30 years of ecological observations and housing market transaction records from Oneida Lake, New York, we use a hybrid spatial difference-in-differences and repeated-sales design to disentangle the capitalized values of water clarity and recreational fish abundance. We find that excluding fish populations yields counterintuitive estimates for water clarity, but that controlling for the fishery resolves this bias. We also find that fish abundance capitalizes along the lines of angler utility rather than raw biological biomass, where a one-standard-deviation increase in walleye abundance generates approximately $70 million in aggregate property capitalization. Finally, we find a divergence in spatial amenity footprints, with water clarity capitalizing up to four kilometers inland, whereas the value of fish abundance is capitalized predominantly into lakefront property values. These results highlight the need to consider the role of nutrient concentrations in the joint production of aesthetic and recreational amenities to support multi-use freshwater lake management.

Key Words: Non-market Valuation; Hedonic Model; Water Quality; Fish Populations; Omitted-variable Bias; Spatial difference-in-differences

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1. Introduction

Freshwater ecosystems provide multiple societal benefits through ecosystem services, notably aesthetic amenities and recreational angling (Keeler et al., 2012; Guerry et al. 2015). Evaluating the economic value of such freshwater attributes is essential for evidence-based regulatory analyses and lake management decision making (Moore et al. 2023; Irwin et al., 2018). Yet, policymakers frequently confront a fundamental tension driven by the biophysical realities of nutrient loading. Nutrients such as nitrogen and phosphorus act as a shared ecosystem input that drives competing economic goods: visual clarity and fishery productivity. While low nutrient levels yield the pristine water clarity that homeowners value, they inherently limit the biological productivity required to sustain robust game fish populations. Moderate nutrient inflows degrade visual clarity by stimulating phytoplankton growth, but enhance the abundance of recreational fish, up to a tipping point where extreme eutrophication degrades ecosystem functions, sometimes creating hypoxic ("dead") zones (Ogelsby 1977, Nixon 1982, Scavia et al., 2024; Abbott et al. 2022). These opposing biological dynamics translate directly into economic trade-offs that cannot be captured by valuations of surface waters that focus on a single dimension of use value. Consequently, as Keiser et al. (2019) note, overlooking fisheries in hedonic models of water quality poses a challenge for the benefit-cost analysis of environmental regulations.

The economic literature to date has largely valued these services independently using differing methodological approaches. The hedonic property price model (hereinafter, the hedonic model) is widely used to quantify the economic value of surface water quality, typically proxying aesthetic amenities via water clarity measures (Secchi disk readings) or biochemical characteristics like chlorophyll-a concentrations or nutrient loadings (Boyle et al., 1999; Mamun et al., 2023; Poor et al., 2007; Weng et al., 2020; Wolf et al., 2021). However, as Melstrom et al.

(2023) noted, these applications typically omit measures of fishing quality, often due to data availability constraints. Conversely, recreational demand models are traditionally used to value angling (Timmons and Murdock, 2007). Even recent structural advancements that jointly estimate hedonic and recreational demand models for a single water quality attribute continue to overlook the capitalized value of the fishery itself (Phaneuf et al., 2008; Kuwayama et al. 2022; Melstrom et al. 2022). Because water clarity and fishery productivity are inextricably linked within the ecosystem, omitting fishing quality introduces classic omitted variable bias into the valuation of water quality.

This study bridges these lines of inquiry to explicitly incorporate both lake water quality and fish abundance into a unified hedonic modeling framework. This approach allows us to disentangle the complex and competing effects of water clarity and recreational fish abundance that are bundled in standard reduced-form estimates that rely on a single water quality metric. We start with a stylized framework that decomposes the marginal willingness to pay for lake amenities into an aesthetic channel and a biological productivity channel. Because water clarity and fish abundance are biophysically linked through nutrient levels, estimating the value of water quality without accounting for the fishery forces the aesthetic coefficient to absorb the unobserved value of the biological stock. We show that the direction of this omitted variable bias depends on the lake's trophic state. In eutrophic systems, where excess nutrients cause hypoxia and the collapse of game fish populations, omitting the fishery leads to an overestimate of aesthetic damages, as the water quality parameter implicitly absorbs the collapse of the recreational amenity. In mesotrophic systems where moderate nutrients degrade clarity but sustain booming sport fisheries, the bias reverses. The aesthetic effect of lower water clarity is

offset by the implicit premium associated with a healthy fishery, which attenuates the estimated value of water quality.

To empirically test this effect and isolate the value of both amenities, we require an empirical setting with three characteristics: a mesotrophic (i.e., moderately nutrient-enriched) ecosystem exhibiting temporal variation in water clarity and fish abundance, a long panel of local housing transactions, and a mechanism to convey information about fish populations to homeowners. Oneida Lake, New York State's largest inland freshwater lake, provides the ideal setting. This mesotrophic fishery is renowned for its walleye population, allowing us to explore the trade-offs between water clarity and fishery productivity. Furthermore, Oneida Lake benefits from a rich record of ecological monitoring for both fish populations and water quality via the Cornell Biological Field Station. This scientific data is then distilled and conveyed to the surrounding community via the Oneida Lake Association (OLA). With a highly active membership base and strong local ties, the OLA disseminates these ecological updates through annual meetings, newsletters, and digital platforms, effectively making the broader public and prospective homebuyers aware of fluctuations in the lake's fishery.

Combining 30 years (1990-2019) of ecological observation and housing market transaction data, our empirical strategy employs a hybrid design that combines a repeated-sales hedonic model with a ring-based spatial difference-in-differences model (Banzhaf, 2021; Bishop et al., 2020; Butts, 2023). Specifically, we interact the temporal variability of ecological state vectors in water clarity and fish abundance with a series of discrete distance rings extending from lakefront up to 4km inland. By comparing "treated" properties with direct lake access to the control group of inland properties conditioning on property fixed effects, our hybrid design isolates the capitalization of local ecological and aesthetic characteristics from other property-

level heterogeneities as well as shared local and regional shocks. Identification relies on a local common trend assumption, the validity of which we test by estimating non-parametric treatment effect curves across spatial gradients, as suggested by Butts (2023).

Our empirical results reveal four important findings. First, excluding fish abundance leads to omitted variable bias in estimating the capitalization effects of water clarity. When fish abundance is omitted, the results for water clarity are statistically insignificant and counterintuitive: the effect of Secchi depth is statistically insignificant, whereas higher total phosphorus (TP) levels increase property values. Controlling for fish abundance yields water clarity coefficients with the expected sign and significance. Second, fishing quality capitalizes into local housing markets in a way that is consistent with angler preferences. Enhanced walleye abundance, the most sought-after gamefish species in the region, carries substantial housing market premiums; a one-standard-deviation increase in the walleye population raises lakefront property-value premiums by 16%. In contrast, greater abundance of secondary species like yellow perch exhibits a negative premium, likely because greater abundance of these fish reduces the catchability of the target trophy species, thereby operating as a recreational disamenity to the angler. Third, our non-parametric spatial spillover analysis reveals a divergence in the spatial footprint of these bundled amenities. Water clarity operates as a diffuse local public good, with capitalization effects extending up to four kilometers inland. In contrast, the housing market premium for fishing quality functions effectively as an excludable club good, with economic benefits strictly confined to lakefront parcels where homeowners possess direct angling access via private docks. Because of that, our observed omitted variable bias is also strictly concentrated among lakefront properties and dissipating entirely beyond the second row. Finally, our back-of-the-envelope analysis suggests that these localized fishery premiums represent a substantial and

previously unmeasured municipal tax base. A one-standard-deviation increase in the walleye population generates $70 million in aggregate property value, yielding over $2.3 million in annual local property tax revenues.

2. Study Area and Data

2.1 Study Area

We study Oneida Lake in the State of New York, USA (Figure 1) to explore the relationships between fish abundance, lake water quality and property values. Long-term data on fish abundance, spatial differences in lake water quality, and real-estate transactions in a mesotrophic setting make Oneida Lake an ideal location to explore the relationships between water quality and fisheries on property values.

Oneida Lake is the largest lake entirely within the borders of the State of New York, with a surface area of 207 km^2, average depth of 6.8 meter, and 54.7 miles (88 km) of shorelines (Rudstam et al. 2016a). Covering four counties in New York (Madison, Onondaga, Oswego, and Oneida), Oneida Lake real estate ranks in the top ten lake home and lake lot markets in the state.[1] Oneida Lake has a diverse fish community and a long history of recreational fishing (VanDeValk et al. 2016). Walleye is the most sought-after species by anglers: during the open water season, anglers typically target walleye on 60% to 70% of their trips, along with largemouth and smallmouth bass (VanDeValk et al. 2016).[2] The lake and its fish are the concern of an active lake and angler organization, the Oneida Lake Association (OLA), with between 2,000-3,000

[1] Source: https://www.lakehomes.com/new-york/oneida-lake. Retrieved 3/3/2025.
[2] Source: https://dec.ny.gov/things-to-do/freshwater-fishing/places-to-fish/statewide-opportunities/walleye

members and strong ties with the local community. The OLA provides multiple opportunities for information dissemination, such as updates on water quality and fishery conditions at their annual meeting presented by scientists from the Cornell Biological Field Station and through newsletters, a web page, and a Facebook page. These communication channels effectively makes the public aware of changes in fish populations.[3] The OLA is active and engaged, as evidenced by receiving the American Fisheries Society's highest award given to an organization involved in fish conservation in North America in 2024.[4]

2.2 Water Quality and Fisheries Data

We follow a long literature valuing the effects of ecosystem attributes on nearby housing markets for freshwater lakes (Gibbs et al., 2002; Kuwayama et al., 2022; Michael et al., 2000; Weng et al., 2020; Wolf et al., 2021). Our analysis uses observational data from Cornell Biological Field Station (CBFS), which provides long-term monitoring for limnological and ecological conditions in Oneida Lake (Rudstam et al., 2016b). During our study period, water quality was monitored weekly at four sampling sites from May to October. Panel A of Figure 2 shows water quality sampling locations. Fish abundance was monitored weekly over the same period with rotational sampling at over a dozen sampling locations (Panel B of Figure 2). Both data are presented in annual reports (Van De Valk et al. 2024) and published in an on-line data repository (Rudstam and Almeida 2023).

Our analysis considers three water quality indicators: Secchi depth readings (hereafter referred to as "Secchi"), total phosphorus (hereafter referred to as "TP"), and water temperature.

[3] OLA newsletters and bulletins can be found on the OLS website, dating back to Spring 1991: https://www.oneidalakeassociation.org.
[4] See American Fisheries Society's announcement: https://fisheries.org/about/awards-recognition/call-for-award-nominations/presidentsfisheryconservationaward/

Secchi is a visual measure of water clarity. Studies have shown that an increase in Secchi is associated with higher lakefront housing values, as lake homeowners are willing to pay more for improved water clarity (Walsh & Milon, 2016). TP is important because the lake is phosphorus-limited, meaning that changes in phosphorus loads are the primary driver of increased phytoplankton growth (Volponi et al., 2022). Increased phytoplankton growth reduces water clarity, but contributes to the growth of fish populations, including walleye. Water temperature is included to account for its impact on biomass growth and recreational activities. We average each water quality indicator across summer months annually and by sampling site.

To measure fish abundance, we use gill net catch data as an index of fish abundance and trends over time. Every year, gill nets were set following a fixed route during summer months, from the first Monday in June through mid-September. These standard gill net surveys have been conducted each year since 1957 and are highly correlated with mark-recapture estimates for walleye and yellow perch (Rudstam et al. 2016c). These data are used to index different fish species each year and analyzed in multiple studies of fish population and ecosystem dynamics (e.g. Rudstam et al. 2004, 2016c). Details of fishing methods are in these publications and in the online database (Rudstam and Almeida 2023).

For each year, we aggregate weekly sampling data from summer months across space and time into a single metric of average gillnet catches for each species. While there are sizeable spatial variations across sampling locations (see Vidal et al. 2017), we keep the annual trends in gillnet catches for two reasons.[5] First, these spatial variations are driven by either microhabitats within Oneida Lake, or by the temporal pattern within summer months since each site is sampled

[5] For example, Lewis Point, on the Southeast side of the lake, sees an average gill net catch of 40.5 walleyes and 11.1 white perches. Darkin Shoal, on the North side, averages 7.6 walleyes and 72.6 white perches.

roughly the same time each year. Secondly, unlike water quality amenities, anglers can move their boats to different locations on the lake to catch their target species. We consider four fish species that are important to anglers in Oneida Lake: walleye (*Sander vitreus*), smallmouth bass (*Micropterus dolomieu*), yellow perch (*Perca flavescens*), and white perch (*Morone americana*). To facilitate comparison across different fish species, we standardize species abundance by creating z-scores (with a mean of zero and a variance of one) for gill-net catches of each species over time.

Table 1 presents summary statistics for water quality and fish abundance for our sample period from 1990-2019. Across years, the average Secchi depth was 3.6 meters (standard deviation (SD=0.5 meter), average TP concentrations were 21.9 ug/L (SD=4.8 ug/L), and the average water temperature was 19.78 degrees Celsius (SD=0.72 °C). For fish abundance, average gillnet catches (number of fish per net) were as follows: walleye 19.1 (SD = 6.9), smallmouth bass 2.7 (SD = 1.0), yellow perch 42.4 (SD = 11.8), and white perch 25.5 (SD = 12.9). Time series changes of water quality indicators, fish abundance, and their correlations are presented in Appendix Figures 2 and 3.

2.3 Property Data

We obtained property data from Estated, which includes information on the structure, deeds, parcels, transaction history, as well as the geographical location of each property.[6] We extracted arms-length real estate transactions for single-family homes from 1990 to 2019 within the four counties (Madison, Onondaga, Oswego, and Oneida) of the Oneida Lake catchment area.[7] We

[6] Estated now becomes a part of ATTOM Data Inc.. We acquired transaction data up until March 2020.
[7] An arm's length transaction refers to the property transactions in which buyers and sellers act independent without one party influencing the other.

windsorized (dropped) the top and bottom 2% of all sales records, or any sales records below $33,614 or above $387,958 (in 2012 dollars), yielding a final dataset containing 184,744 sales records. The average sales price for our samples is $129,890.

A priori, we anticipate that the impact of fish abundance and water quality differ between lakefront and non-lakefront properties since lakefront homeowners likely pay premiums for both water quality and recreational fishing opportunities while non-lakefront properties do not have these adjacency premiums (Zhang et al. 2022). We visually identified lakefront and adjacent lakefront (second row) properties using Google Maps and satellite imagery from Google Earth. Lakefront properties are those that have direct access to the lake; second row properties are those that need to cross an additional property and/or road to access the lake (Figure 1).[8] We calculated the Euclidian distance for each property to the Oneida shoreline based on its location.

Neighborhood attributes also impact housing prices. In our analysis, we include a suite of community factors and neighborhood characteristics, such as socio-demographic characteristics, school districts, distance to the nearest Walmart, and distance to the nearest post office. Data on age, race, and education are obtained from the U.S. Census and summarized at the census-tract level. School district boundaries are obtained from the New York State GIS Clearinghouse.[9] Euclidian distance to nearest Walmart and nearest post office serves as a proxy for the convenience of property locations to frequent homeowner activities.[10]

Table 2 provides summary statistics for the housing transaction data. Compared to non-lakefront properties, lakefront homes are more expensive, newer, have fewer bedrooms, and are

[8] Some second-row properties may have owned or deeded access to the lake but we do not have data to identify such rights. Second-row properties belong to the non-lakefront properties categories.
[9] Available at: https://gis.ny.gov/gisdata/inventories/details.cfm?DSID=1326. Retrieved 5/3/2021.
[10] Location of Walmart stores are available at https://walmart-open-data-walmarttech.opendata.arcgis.com/. Post office locations are obtained from the USPS at https://postalpro.usps.com/gis/po-location-icd.

located further from shopping areas. Additionally, residents in lakefront communities are relatively older, with a lower percentage of non-white residents and lower education levels. There is no statistically significant difference in square footage between lakefront and non-lakefront properties.

Linking fish abundance and water quality data with property sales data is essential to the estimation of hedonic model. Following Weng et al. (2020), we merge fish abundance and water quality data with property sales data based on sale year. If a sale happened between January 1st and June 30th, we use the prior sale year's fish abundance and water quality data since most recreational activities occur in summer. If the sale happened between July 1st and December 31st, we use the sale year fish abundance and water quality data.

3. Empirical Strategy

3.1 Decomposing aesthetic vs. angling amenities

We base our analysis in the hedonic property price model (Rosen, 1974), where the equilibrium price of a residential property is determined by a bundle of its attributes. In the context of freshwater lakes, standard hedonic applications typically model ecosystem services as a scalar measurement of water quality, for example Secchi disk readings, water quality indices, or nutrient levels. A disadvantage of this approach is that it implicitly assumes that water quality maps to a single dimension of household utility. We argue that this specification is incomplete because nutrient levels jointly produce two distinct goods with competing effects on property values: visual amenities (water clarity) and fishery productivity (abundance of recreational game

fish). By treating these distinct services as a single variable, reduced-form models risk conflating the value derived from each.

To formalize the intuition, let the price of property i at time t, P_{it}, be a function of structural characteristics $\boldsymbol{Z}_{it}$ (e.g., lot size, number of bedrooms and bathrooms) and a vector of environmental amenities $\boldsymbol{E}_{it}$, such that,

$$\mathrm{P_{it}} = \mathrm{P}(\mathbf{Z}_{\mathrm{it}}, \mathbf{E}_{\mathrm{it}}) \tag{1}$$

In standard applications, $\boldsymbol{E}_{it}$ is often treated as a scalar measurement of water quality, q_{it}, proxied by either water clarity measures, e.g., Secchi disk readings, or a measure of the underlying nutrient levels (N_{it}), e.g., total nitrogen (TN) or total phosphorous (TP). However, this scalar treatment overlooks more complex pathways of how water related amenities enter household utility functions and thus equilibrium housing prices. Specifically, nutrient loadings act as a common input for two distinct ecosystem services: visual amenities (A) and recreational fishery productivity (F), both of which depend on nutrient levels.

We decompose environmental amenities as $\mathbf{E}_{\mathrm{it}} = \{\mathrm{A}(\mathrm{N}_{\mathrm{it}}), \mathrm{F}(\mathrm{N}_{\mathrm{it}})\}$. The household maximization problem yields a price function that reflects the marginal willingness to pay (MWTP) for these attributes. Totally differentiating the equilibrium price function with respect to the underlying nutrient load (N) reveals the structural decomposition of value:

$$\frac{\mathrm{dP}}{\mathrm{dN}} = \underbrace{\frac{\partial \mathrm{P}}{\partial \mathrm{A}}\frac{\partial \mathrm{A}}{\partial \mathrm{N}}}_{\text{Amenity Channel (-)}} + \underbrace{\frac{\partial \mathrm{P}}{\partial \mathrm{F}}\frac{\partial \mathrm{F}}{\partial \mathrm{N}}}_{\text{Productivity Channel (?)}} \tag{2}$$

Equation (2) formalizes the two competing economic forces triggered by a change in nutrient levels. The first term, the amenity channel, captures the capitalization of visual aesthetics. As nutrient level N increases, water clarity (A) deteriorates ($\frac{\partial A}{\partial N} < 0$). Since households have a

positive marginal utility for visual amenities ($\frac{\partial P}{\partial A} > 0$), increased nutrient levels reduce property values. The second term captures the housing market capitalization of recreational fishery productivity. Changes in nutrients alter the productivity along the food web (F), which in turn affects the recreational value derived from the lake ($\frac{\partial P}{\partial F} > 0$). The direction of this effect depends on the sign of $\frac{\partial F}{\partial N}$, which we argue is a function of the lake's trophic status.

In eutrophic or oxygen-limited systems (e.g., the Western Basin of Lake Erie), the relationship between nutrients and fishery health is negative ($\frac{\partial F}{\partial N} < 0$) as excessive nutrient levels lead to harmful algal blooms (HABs) and bacterial decomposition, reducing dissolved oxygen (DO) below the thresholds required for game fish survival (hypoxia). In this regime, the amenity and productivity channels are complementary: nutrient pollution simultaneously degrades the visual landscape and the recreational fishery. Conversely, in mesotrophic systems such as Oneida Lake, moderate nutrient loading stimulates phytoplankton growth, which cascades up the trophic web to support higher densities of zooplankton and prey fish, ultimately enhancing the stock of recreational game fish ($\frac{\partial F}{\partial N} > 0$). Here, the amenity and productivity channels are competing. An increase in nutrients degrades visual clarity but enhances fishery abundance, creating a trade-off.

Now consider a typical reduced-form regression of housing price P_{it} on a biophysical indicator, N_{it}, without including fish abundance. The resulting estimate of the marginal effect of N on P, $\frac{dP}{dN}$, will recover the total derivative of the bundle of ecosystem services, i.e. the aggregate capitalization value. In eutrophic systems where nutrients degrade both aesthetic amenity and fishery productivity, the reduced form property premium will overstate the amenity

effect. In mesotrophic systems where the amenity and productivity channels operate in opposing directions, the reduced-form property premium captures a net effect where the capitalization of improved clarity is offset by the capitalization of reduced fishery abundance. This has the effect of dampening the estimated property premium associated with water quality. Empirically, we anticipate that this effect will drive standard reduced form estimates toward zero or even induce sign reversals, leading to the counterintuitive (and erroneous) conclusion that homebuyers place no value or negative value on water clarity in mesotrophic lakes.

3.2 Identification Strategy

To isolate the capitalization effects of lake ecosystem services, we employ a hybrid design that combines a repeated sales hedonic model with a ring-based design similar to a spatial difference-in-differences model (Butts, 2023). While the repeated sales model leverages within-property variation to absorb time-invariant unobserved heterogeneities such as property characteristics or school-zone quality, the ring-based design further leverages the fact that both aesthetic amenities and access to recreational angling are highly localized to the immediate areas next to the lake and their importance decays quickly over space. By comparing "treated" properties within a proximity ring of the lake against "control" properties outside of the ring, this approach isolates the differential capitalization of ecological fluctuation from local and regional level shocks driving the housing market.

We start by categorizing properties based on their proximity to the lake. Let L_i be a binary indicator equal to one if property i is treated, i.e., has direct access to the amenity bundle, and zero otherwise. The treatment group consists of properties closer to the lake, i.e., the lakefront properties, whereas properties further inland constitute the "control group," which share the same local public goods and macro shocks but lack direct access to lake ecosystem

services. We detail methods to empirically select treatment buffer rings later in the section. The "treatment intensity" is defined by the annual variation in the ecological state vectors, WQ_{ikt} for a vector of K water clarity indicators, and $Species_{jt}$ for a vector of J fishery abundance indicators.

A critical feature of our treatment definition is the recognition that "treatment" is defined not just by proximity, but by the nature of the ecosystem service. While we start by using lakefront properties as the primary treatment group for both water quality and fish abundance attributes, we acknowledge that the spatial footprint of their capitalization likely differs. Angling is an access-dependent activity, implying a potentially sharp treatment boundary at the water's edge. In contrast, water quality amenities may project further inland beyond properties with immediate access to the lake (Kuwayama et al., 2022; Wolf et al., 2021).

Our preferred specification takes the form:

$$\log(\text{SalePrice})_{it} = \sum_k \beta_k(\text{WQ}_{ikt}) + \sum_k \gamma_k(\text{L}_i \times \text{WQ}_{ikt}) + \sum_j \theta_j\left(\text{L}_i \times \text{Species}_{jt}\right) + \text{Control}_{it} + \lambda_i + f(t) + \epsilon_{it} \quad (3)$$

where $log(SalePrice)_{it}$ denotes the logarithm of the sales price (inflation-adjusted) for property i sold in year t; L_i is a dummy variable that equals 1 if the property is lakefront, and 0 otherwise; WQ_{ikt} represents the average of water quality variable k in year t for property i, which includes Secchi disk depth, total phosphorus (TP), and water temperature. $Species_{jt}$ represents the abundance of fish species j in year t, where j={walleye, smallmouth bass, yellow perch, white perch}.[11] $Control_{it}$ refers to individual and time-varying characteristics, including property age. Following best practices suggested by Bishop (2020), we include fixed effects at the individual

[11] We include the level of water quality (WQ_{ikt}) since it has both spatial and temporal variation. The level of fish abundance (Species_{jt}) is omitted from the model because it has only temporal variation and thus will be absorbed by the time trend.

house level, λ_i, and flexible controls for time-trends $f(t)$ to control for individual unobserved heterogeneity and time trends.[12] ϵ_{it} is the idiosyncratic error term.

The coefficients of interest, γ_k and θ_j, operate as continuous difference-in-differences (DID) parameters. By interacting the temporal variation in ecological amenities (the "treatment intensity") with the spatial indicator for the lakefront (dummy for "treatment ring"), these coefficients capture the differential percentage price of treated properties relative to their inland neighbors in response to fluctuations in the lake's ecological state. For γ_k, the coefficients of water quality metrics, we expect the impacts on lakefront property premiums to vary depending on the specific water quality measurement: the coefficient for Secchi is expected to be positive; TP to be negative as it drives algae growth, harming water quality; and mean water temperature to be positive as warmer waters improve water-based recreational activities such as swimming (e.g., Weng et al. 2020; Wolf et al. 2021).[13] Regarding the coefficients of fish abundance (θ_j), we anticipate positive capitalization effects for species targeted by anglers, with the magnitude varying by species. The most significant effects are expected for regionally prominent game fish species, particularly walleye. The impact of other species likely depends on their recreational value to residents of the region.

Identification of our parameters of interest relies on a local common trends assumption as documented in Butts (2023). This framework addresses two distinct sources of potential bias. First, the inclusion of property fixed effects (λ_i) controls for time-invariant unobserved heterogeneities (the "levels"). Commonly known as the repeated-sales in hedonic studies, this

[12] We interpret our main result using 5-degree polynomial time trend. We also test for 3-degree polynomial time trends, and the estimated effects are similar. We cannot include time fixed-effects directly in the model as it will soak up time-series variation in fishery population changes.
[13] We include TP to capture non-visual eutrophication disamenities, such as odor or contact health risks, that are not perfectly collinear with clarity.

strategy differences out property-level unobservables, which has been shown to be important in water quality applications (Kuwayama et al., 2022). A potential concern is that restricting the sample to repeated sales could introduce selection bias, as noted by Freeman et al. (2014). To address this, we report the sample mean differences between repeatedly sold properties and the full sample in Table 3 and map the locations of repeatedly sold properties in our sample (Appendix Figure 1). Overall, repeatedly sold properties tend to be slightly more expensive, smaller, and relatively newer. However, these differences are not substantial, and a visual comparison of Figure 1 (entire sample) and Appendix Figure 1 (repeated sales subset) shows no significant differences in geographic distribution between the two samples. Therefore, the repeated sales model serves as a representative proxy for the entire sample and can be considered an effective method by addressing time-constant omitted variable biases (Bishop et al., 2020).

Second, the distinction between treated properties closer to the lake and the control group further inland provides a counterfactual for the dynamic evolution of local and regional level housing prices (the "trends"). We assume that, in the absence of biological fluctuations in the fishery, price appreciation for treated properties would follow the same trajectory as the inland control properties, sharing the capitalization of school quality, local tax regimes, and macroeconomic shocks. Apart from the suggestive test described below, a violation of this assumption would require an omitted variable that is both spatially concentrated on the shoreline and temporally correlated with variations in water chemistry and/or fish biomass. This is unlikely: the abundance of game fish is driven by complex biophysical factors, such as water temperature, food web dynamics, and stocking histories, that are orthogonal to the socioeconomic factors typically driving differential neighborhood demand. It is unlikely that an

unobserved shock would affect only the lakefront properties but not properties further away, while coincidentally mimic the temporal variation of walleye population.

While the local common trends assumption is not directly testable, we can assess its validity by examining the spatial decay of the capitalization effect. Following Butts (2023), we estimate a flexible, non-parametric "treatment effect curve" by interacting species abundance with a series of distance bins extending outward from the lake. This procedure allows us to empirically verify the spatial extent of the amenity and check for "zero effects" in our control ring. If the local common trend assumption holds, we should observe significant capitalization effects concentrated within the treatment zone, and estimates becoming indistinguishable from zero at some distance further outside of the treatment ring. Results presented in Section 4.2 confirm this pattern.

Finally, to empirically quantify the attenuation bias theorized in Section 3.1, we first estimate a baseline standard hedonic specification that omits the fishery vector, which mimics the conventional approach in the hedonic water quality literature. We then compare these estimates to our preferred specification that includes both water quality and fish abundance variables to isolate the "masking effect" that arises when omitting fish abundance.

4. Results

4.1 Capitalization Effects of Water Quality

We first present our baseline estimates of the impact of water quality on lakefront house premiums in Table 4. Column (2) in Table 4 reports estimated coefficients on the impact of water quality measurements on lakefront property values without controlling for fish abundance,

effectively replicating the standard specification found in the literature. Column (3) presents our preferred specification that isolates the amenity channel by explicitly controlling for the fishery stock. We include results using spatially robust Conley (1999) standard errors in Appendix Table 1.

Focusing first on water clarity (Secchi depth), the specification omitting fish abundance (Column 2) yields a statistically insignificant and negative coefficient. This is likely because water clarity is negatively correlated with walleye abundance. Column (3) reveals that this result is an artifact of omitted variable bias: once fish abundance is controlled for, the water clarity coefficient becomes positive and statistically significant. This sign reversal confirms the "masking effect" predicted in Equation (2) for a mesotrophic lake: because walleye abundance is negatively correlated with water clarity (see Appendix Figure 3), the unobserved biomass productivity premium offsets the amenity effect, driving the reduced-form estimate toward zero.[14]

The results for total phosphorus (TP) and water temperature further corroborate this mechanism. In the model omitting fish abundance (Column 2 of Table 4), TP exhibits a positive and statistically significant effect on property values. This implies, counterintuitively, that homebuyers prefer nutrient-rich water. When we control for fish abundance in Column (3), the TP coefficient drops in magnitude and becomes statistically indistinguishable from zero. This result captures the productivity channel: higher nutrients stimulate the food web, supporting the walleye population. Similarly, the estimate for water temperature in Column 2 is large and

[14] Secchi disk reading is also positively correlated with smallmouth bass and negatively correlated with the abundance of white perch. Though walleye has the strongest correlation with Secchi disk reading amongst all species. The readers are referred to Figure A5 for a full correlation matrix between fish population and water quality indicators.

positive, conflating the preference for warm-water swimming with the biological dependence of fish growth on temperature. Controlling for the stock of fish reduces this coefficient, isolating the residual amenity value of the water temperature itself.

Beyond the lakefront, we analyze the spatial extent of these capitalization effects by estimating the non-parametric treatment effect curve, which allows us to visualize the diffusion of the amenity without imposing arbitrary spatial boundaries. Results are shown in Figure 3. For water clarity (Secchi), the preferred specification (red series) reveals a smooth spatial decay: the capitalization effect is largest for lakefront and second-row properties and declines gradually with distance. Notably, the coefficient remains statistically significant up to 4 km inland, although the magnitude is much smaller compared to the treatment effect for lakefront properties. This persistence suggests that water quality functions as a diffuse local public good, with its impacts capitalized into properties further away from the lake, potentially through visual aesthetics, neighborhood reputation, and perceived environmental health. The smooth decline of the treatment effect curve validates the local common trends assumption; the fact that the effect eventually dissipates confirms that our inland control group is untreated by the amenity. This underscores the value of the flexible ring-based design (Butts, 2023), as a rigid single treatment buffer would likely misrepresent the diffuse nature of these spillovers.

The spatial patterns for secondary water quality metrics are less consistent. For total phosphorus (TP), we find positive capitalization only for properties located 500–4,000 meters from the lake, with no significant effect in the immediate vicinity.[15] For average temperature, we

[15] It could be that properties located farther away have higher values and contribute more to phosphorus loading due to their fertilized lawns and agricultural lands, whereas nearby properties may adopt management practices that limit phosphorus runoff.

find a statistically significant capitalization effect for second-row properties but not at other distances.

More importantly, the spatial heterogeneity in Figure 3 provides an empirical validation of the competing effects of nutrients on water clarity and fish populations, described in Section 3.1. By comparing the estimates without fish abundance (blue triangles) with the specification with fish abundance included (red dots), we can trace the spatial footprint of the masking effect. The divergence between the two specifications is concentrated in the immediate proximity of the lake: the gap between the two estimates is largest for lakefront properties, but vanishes almost entirely for properties beyond the second row, where the estimates from both models converge. This spatial pattern suggests that the capitalization for fish abundance is limited to lakefront properties, with negligible treatment effects beyond the second row. Consequently, the bias generated by omitting the fishery disproportionately applies to lakefront properties. The observation that the masking effect disappears beyond lakefront properties, where the capitalization of fish abundance drops to zero, provides robust confirmation that the fishery was indeed the omitted variable driving the attenuation in the model excluding fish abundance.

4.2 Capitalization Effects of Fish Abundance

To isolate the value of the fishery, we conduct a parallel analysis examining the impact of fish abundance on lakefront house premiums, both with and without controls for water quality. The estimation results are presented in Table 4. We consider two specifications: column (1) excludes

water quality controls, while column (3) includes them. We include results using Conley (1999) standard errors as a robustness check in Appendix Table 1.

Focusing on our preferred specification (Column 3), we find a robust and economically significant capitalization of the lake's primary game fish. A one-standard-deviation increase in walleye abundance raises the lakefront premium by 16.5%. This result aligns with expectations, as walleye is the most sought-after species by anglers at Oneida Lake. We also find a positive and statistically significant effect of white perch abundance for lakefront properties. Interestingly, unlike the water quality estimates in Section 4.1, which were highly sensitive to model specification, the fishery estimates are robust to the inclusion or exclusion of water quality variables. In other words, while omitting the fishery biases water quality estimates, omitting water quality has negligible impact on fishery estimates. This suggests that in Oneida, the fishery is the dominant driver of capitalized values for lakefront homes.

The results also reveal heterogeneity across species, reflecting angler preferences. While walleye and white perch yield positive effects, the estimated coefficients for yellow perch and smallmouth bass are negative. For yellow perch, this is due to the "forage-predator" dynamics of Oneida Lake. While walleye is the target species for 60-70% of angler trips (VanDeValk et al., 2016), and yellow perch serves primarily as forage rather than a target species for most anglers.[16] Ecological studies in this system have documented an inverse relationship between prey density and walleye catchability: when yellow perch abundance is high, walleye are satiated and less

[16] According to New York Angler's Survey, only 4% of the anglers fish primarily for yellow perch in Oswego and Oneida counties, the two counties covering the water body of Oneida Lake (NYSDEC, 2019). In contrast, 23-30% of anglers primary fish for walleye in those two counties, topping the angler's preference list.

likely to strike angler lures (VanDeValk et al., 2005).[17] Consequently, a surplus of forage fish functions as a disamenity to the angler, reducing the "catch per unit effort" (CPUE) of the target trophy fish in walleye. For smallmouth bass, the abundance estimates are inherently noisy since they cling to rocky shoals rather than swarming in the open water(Shoup & Ryswyk, 2016), making them harder to be sampled by gill nets.

Finally, we examine the spatial heterogeneity of capitalization for fish abundance in Figure 4. The findings reveal a striking contrast to the capitalization effects of water quality over space. Unlike the diffuse capitalization of water quality, which extends kilometers inland (Figure 3), the capitalization of fish abundance is confined to the immediate shoreline. Focusing on the repeated-sales specification with water quality controls, we find that walleye abundance has a statistically significant marginal effect of 17.3% on lakefront housing premiums, consistent with the baseline model in Table 4. However, the effect declines sharply with distance: the marginal effect drops to 2.6% for second-row properties, 0.6% for properties within 250 meters (but neither lakefront nor second-row), 2.1% for properties 250–500 meters away, and -1.0% for properties 500–1,000 meters away. All other estimated coefficients are statistically insignificant at the 5% level. White perch shows a similar spatial pattern. The capitalization effect on lakefront properties is significant, with a premium of 11.3%, consistent with baseline estimates. For properties farther from the lake, the effects are smaller and generally statistically insignificant, with the exception of properties 1–4 km away, where the estimated premium is 1.4%, much smaller than that for lakefront properties.

[17] Interestingly, this is not the pattern in other walleye fisheries, for example in Escanaba Lake, WI (Newby et al., 2000) or Lake Erie (Isbell & Rawson, 1989). Part of the reason could be that (VanDeValk et al., 2005) draw statistical inferences on aggregated catch rates from six surveys separated by over 50 years.

This discontinuity validates the economic mechanism of the benefit: while water clarity functions as a local public good that capitalizes at the broader neighborhood scale, the capitalization effect of the fishery functions effectively as a club good. The premium captures the exclusive utility of direct lake access (e.g., private docks), a bundled asset that allows homeowners to bypass the travel costs associated with public launch sites. For second row properties and beyond, the housing market premium for the fishery disappears since such access is no longer provided by the property.

4.3 Capitalized Value of Fish Abundance and Associated Economic Impacts

While travel cost models estimate the active use value of the fishery and capture the welfare of anglers who travel to the lake, our hedonic framework isolates a different value: the capitalized value anchored in the local housing market, especially properties with direct lake access. From a natural capital perspective, our estimate represents an asset value independent from public recreational benefits commonly estimated by travel cost studies, in that it is the asset value of direct lake access bundled in home prices (Fenichel, 2024). In fact, standard travel cost surveys will have a difficult time capturing this value since the cost of travel to the lake for lakefront property owners is negligible.

To quantify and benchmark this value, we calculate the total capitalization value (CV) for a species by summing the marginal premiums across all distance bands, weighted by the housing stock density in each band, i.e.

$$CV_j = \sum_{\mathrm{k}} \beta_{jk} \times \mathrm{NP}_k \tag{4}$$

where β_{jk} is the marginal effect of species abundance on property value for each distance band k and NP_k represents the number of properties within that distance band, which we acquire from

assessor records. We sum the capitalization effects for different distance bands to calculate the total capitalized value of the fish species considered for Oneida Lake.

We find economically sizeable capitalization effects, particularly for lakefront properties: the capitalized value of a one-standard-deviation increase in walleye abundance amounts to $70.0 million, of which $64.8 million is for lakefront properties.[18] We predict the annual property tax revenue (TR_j) as a result of changes in species abundance as:

$$TR_j = CV_j \times t \tag{5}$$

where CV_j is defined in equation (4) and the average effective tax rate, *t,* is calculated by dividing the average amount of property tax by the assessed property value, both of which we acquired from assessor records. We find that a one-standard-deviation increase in walleye abundance increases local property tax revenue by $2.38 million per year, of which $2.20 million can be attributed to increased property values for lakefront homes. For each individual homeowner, our estimates suggest that the annual rent premium for a one-standard-deviation increase in the walleye population is around $872 under a 3% interest rate.

To benchmark the relative magnitude of our estimated effects, in Panel B of Table 5, we compare our estimated property tax implications with tax revenues associated with recreational fisheries in the four counties adjacent to Oneida Lake, estimated in the New York Angler's Survey (NYSDEC, 2019). While the angler's survey is the closest estimate we could find, the numbers are overstated for Oneida Lake because it reports tax revenues for all recreational fishery activities within each county, which includes other lakes. For example, Oswego County has prominent alternative fishing sites on the Oswego River (feeding from Oneida Lake through

[18] All monetized values are in 2012 dollars.

the Oneida River) and Lake Ontario and so has much higher tax revenues and mean angler travel distances. In the three other nearby counties, we find that the magnitude of the property tax effects is a sizeable portion of the total tax revenue: in Oswego County, increases in property taxes are comparable to 33% of the total fishing-related tax revenue; in Onondaga County, 36%; in Madison County, 131%. Our exercise highlights the public finance implications of recreational fisheries, not only through direct consumption (for example, dining, lodging, and fishing gear) but also through property market linkages.

5. Conclusion and Discussion

Unbiased and accurate non-market valuation of ecosystem services is essential for effective freshwater conservation and policy analysis. However, empirical valuation is challenging when biophysical processes produce competing amenities. We address this challenge by disentangling the capitalized values of two fundamentally linked services in freshwater lakes: water clarity and recreational fishery productivity. Focusing on a mesotrophic system, we show that traditional hedonic models that rely on a single water quality metric suffer from omitted variable bias — a bias that is resolved after controlling for fish abundance. By isolating the value of angling to lakefront property owners, we recover a broad, positive willingness-to-pay for water clarity that standard models otherwise obscure or even reverse in sign.

Beyond the econometric bias, the magnitude of the capitalized value of fish populations carries important policy implications at the local, state, and national level. At the local level, a one-standard-deviation increase in the walleye population generates a $70 million expansion in aggregate property value. Because local communities depend heavily on property tax revenues to

fund local governments and public schools, these localized fishery premiums represent a massive, previously unmeasured component of the municipal tax base. At the state level, authorities such as the New York State Department of Environmental Conservation are responsible for managing both lake water quality and recreational fisheries. Our results indicate that relying on biased or missing benefit estimates will lead to sub-optimal policy outcomes: nutrient reduction targets that ignore fishery productivity could over-invest in water clarity improvements while under-weighting the recreational costs of those same policies — potentially setting nutrient thresholds that reduce walleye abundance and, with it, the property tax base that funds local schools. Our findings are also important to consider when conducting benefit-cost analysis, which should account for the joint, and sometimes competing, benefits of water clarity and fishery productivity to avoid placing erroneous values on environmental outcomes.

We focused on a single lake in this analysis because of the availability of long-term ecological data on both water quality and fish abundance. Even so, the omitted variable bias identified here is fundamentally tied to the biophysical realities of mesotrophic lakes, making our findings highly relevant to the broad class of moderately nutrient-rich water bodies with managed fisheries across the upper Midwest and northeast United States. A natural extension would be to apply our approach to evaluate the spatial heterogeneity of bundled amenities across diverse lake systems. Future studies can leverage the growing availability of spatially explicit creel surveys maintained by fishery biologists, systematically integrated with large-scale citizen science water quality monitoring datasets, to test the external validity of our findings across varied ecological and institutional settings. Furthermore, integrating these disentangled capitalization estimates into structural bioeconomic models offers a promising avenue for explicitly optimizing upstream land-use and agri-environmental policy. Because lake nutrient

loading is fundamentally driven by non-point source agricultural runoff, linking the downstream value of this amenity bundle to watershed-level land-use decisions would enable economists to better evaluate systemic policy trade-offs — providing a clear framework for assessing the return on investment for agricultural conservation programs and ensuring that nutrient management targets maximize the welfare provided by aesthetic and recreational ecosystem services. Finally, benchmarking our capitalized fishery premiums against estimates from travel cost or stated preference models would provide cross-validation across methodologies and yield a more comprehensive picture of the total non-market value of recreational fish populations — capturing both the asset value embedded in lakefront housing and the active use value accruing to visiting anglers who do not own property on the lake. Ultimately, the central lesson of this paper is that the economic value of a lake cannot be reduced to a single dimension. The full value is bundled, the amenities compete, and accurate measurement requires confronting that complexity directly.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Gemini (version 2.0 Pro) and Claude.ai (version Sonnet 4.6) in order to conduct proofreading. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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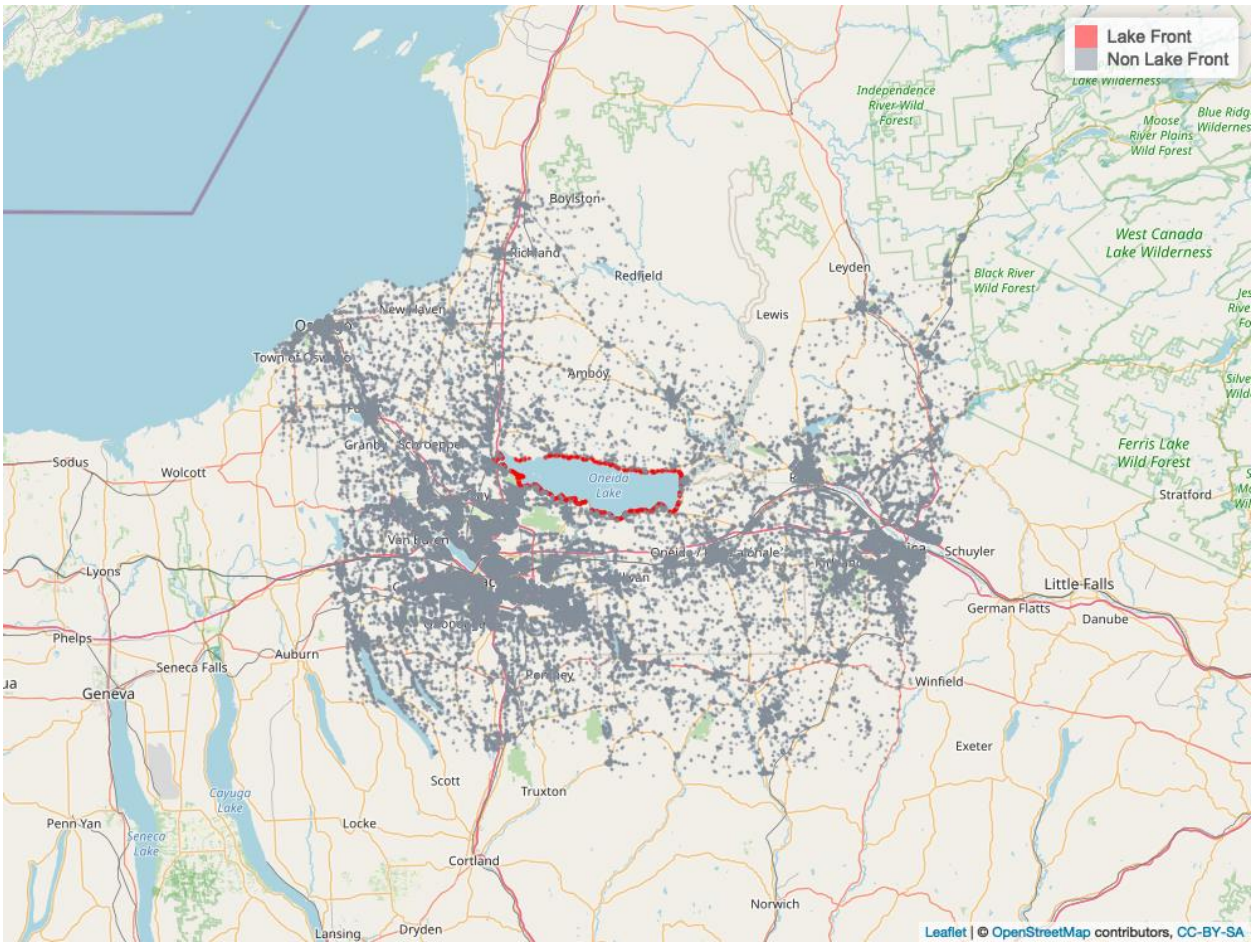


Figure 1: **Property Sales Near Oneida Lake.** Note: Each black dot denotes the location of one property. Red dots denote properties with frontage of the Oneida Lake (lakefront). Grey dots denote non-lakefront properties.

Panel (A): Water quality sampling locations

Panel (B): Gill net sampling locations

Figure 2: **Gill net and water quality sampling locations.** Panel (A) displays water quality sampling locations. Panel (B) displays gill net sampling locations. Each blot dot denotes a sampling location, with its respective names denoted alongside.

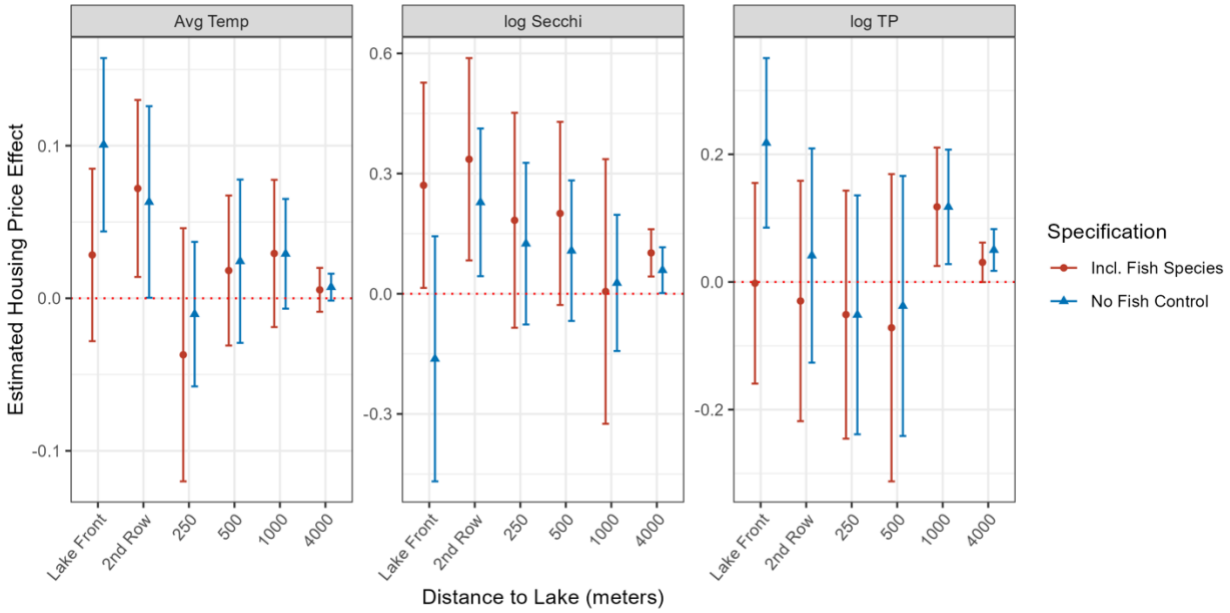


Figure 3: Spatial Spillover for Water Quality Capitalization with or without fishery controls. Note: Panels show coefficient estimates for the capitalization effects of average water temperature (left), the logarithm of Secchi disk reading (middle), and the logarithm of total phosphorus (right), over different proximities to Lake Oneida. Property fixed effects are included in all regressions. Distance intervals include properties that are lakefront, on the second row, within 250 meters of Oneida (but not lakefront or second row), 250-500m, 500-1000m, and 1000-4000m. Error bars denote 95% confidence intervals. Red dots and error bars denote estimated impacts of water quality by including interaction terms between property proximity bands and the four fish species: walleye, smallmouth bass, white perch, and yellow perch; Blue triangles and error bars denote the estimated impacts of water quality without controlling for fish abundance.

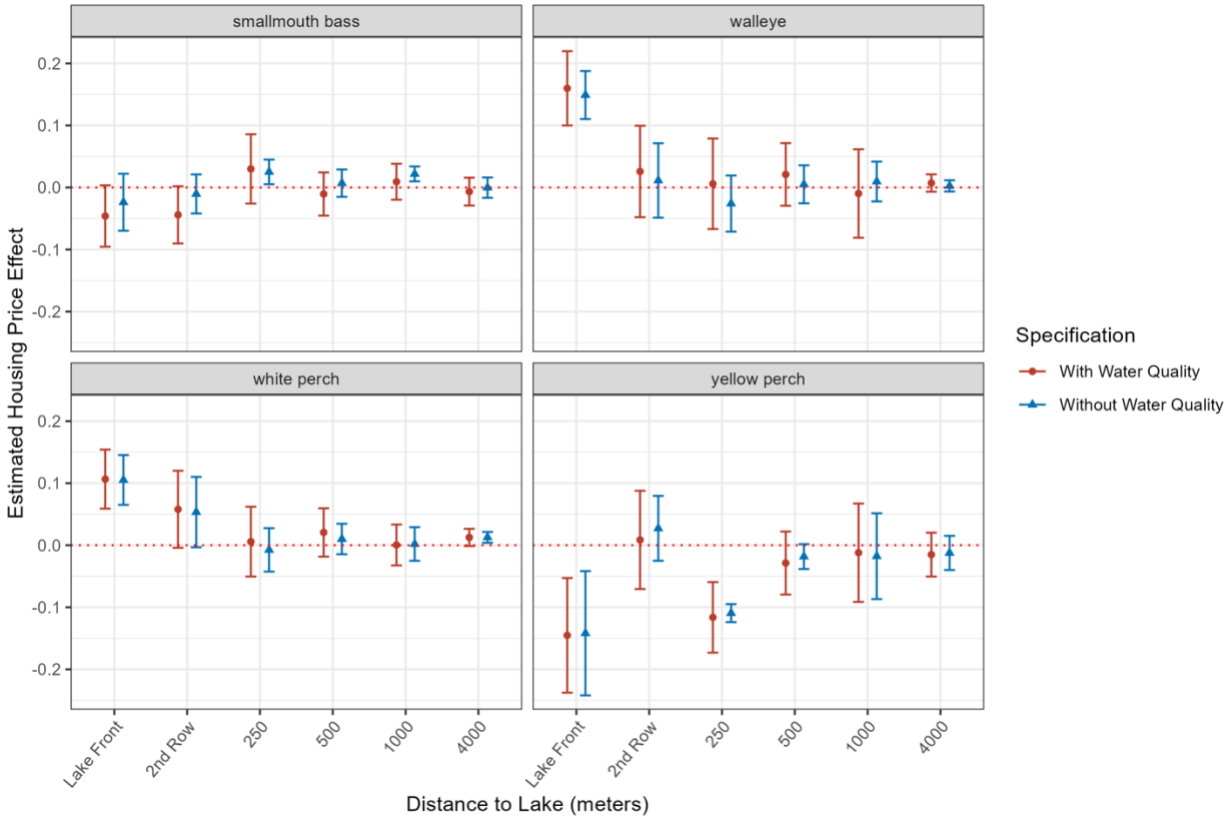


Figure 4: Spatial Spillover for Capitalization of fisheries, with and without water quality controls. Panels plot point estimates of individual species abundance interacted with proximity dummy variables, corresponding to equation (2). Property fixed effects are included in all regressions. Distance intervals include properties that are lakefront, on the second row, within 250 meters of Oneida (but not lakefront or second row), 250-500m, 500-1000m, and 1000-4000m. Control group are properties beyond 4km of the Oneida shoreline and are omitted. Red dots and error bars depict point estimates and 95% confidence intervals from the specification that include water quality variables interacted with proximity dummy variables as controls. Blue triangles and error bars depict point estimates and 95% confidence intervals from the specification that exclude water quality variables as controls.

Table 1: **Summary Statistics of Ecological Variables**

	Mean	SD	Min	Max
	Water Quality Indicators			
Secchi Disk Reading (m)	3.626	0.488	2.371	4.796
Total Phosphorus (µg/L)	21.914	4.822	13.475	35.748
Avg Temp (degree C)	19.775	0.718	17.73	21.41
	Average Gillnet Catches			
Walleye	18.904	6.86	5	31.8
Smallmouth Bass	2.746	1.029	1.133	5
Yellow Perch	42.503	11.643	21.067	70.733
White Perch	25.297	13.077	2.467	51.4

Note: Average gillnet catches are calculated as the average summer gillnet sampling catches for each species within a given year.

Table 2: Summary Statistics of Properties

	Lakefront Properties		Non-lakefront Properties		Difference	Significance
	Mean	SD	Mean	SD		
Real Sales Price ($1,000, in 2012 dollars)	174.063	87.932	132.362	75.055	41.701	***
Distance to Oneida Lake (km)	0.045	0.029	20.868	11.69	-20.823	***
No. Bedrooms	2.794	0.849	3.181	0.741	-0.387	***
No. Bathrooms	1.556	0.67	1.476	0.612	0.08	***
Square Footage	1608.972	662.245	1610.334	607.277	-1.362	
Lot Size (acres)	0.541	0.848	1.824	52.274	-1.283	
Property Age	46.144	31.154	50.879	36.699	-4.735	***
% Senior (age>60)	0.268	0.081	0.254	0.091	0.014	***
% White	0.971	0.027	0.894	0.136	0.077	***
% College and Above	0.529	0.134	0.658	0.144	-0.129	***
Distance to Nearest Walmart (km)	12.428	4.304	7.438	6.174	4.99	***
Distance to Nearest Post Office (km)	3.901	2.422	2.276	1.581	1.625	***
N	1,488		183,256			

Note: Pairwise mean difference test reported between lakefront and non-lakefront properties in column 6, and statistical significance code reported in column 7. ***p<0.01, **p<0.05, *p<0.1.

Table 3: Summary Statistics of Repeatedly Sold Properties.

	Repeatedly Sold Properties		Full Sample		Difference
	Mean	SD	Mean	SD	
Real Sales Price ($1,000, in 2012 dollars)	134.6	75.385	132.698	75.26	1.902***
Distance to Oneida Lake (km)	20.166	11.614	20.7	11.79	-0.534***
No. Bedrooms	3.186	0.731	3.178	0.743	0.008***
No. Bathrooms	1.484	0.613	1.476	0.613	0.008***
Square Footage	1601.434	598.732	1610.323	607.737	-8.889***
Lot Size (acres)	1.392	2.952	1.805	51.891	-0.413
Property Age	50.222	36.023	50.841	36.66	-0.619***
Fraction Senior (age>60)	0.252	0.091	0.254	0.091	-0.002***
Fraction White	0.895	0.131	0.894	0.136	0.001
Fraction College and Above	0.665	0.143	0.657	0.145	0.008***
Distance to Nearest Walmart (km)	7.232	5.91	7.478	6.177	-0.246***
Distance to Nearest Post Office (km)	2.261	1.558	2.289	1.596	-0.028***
# Sales	117,825		184,744		
# Property	46,505		113,424		

Note: Pairwise mean difference test reported between lakefront and non-lakefront properties. ***p<0.01, **p<0.05, *p<0.1.

Table 4: Capitalization Effects of Fish Abundance and Water Quality Indicators

	Dependent Variable: log(sale price)				
	(1)	(2)	(3)	Difference: (1) - (3)	Difference: (2) - (3)
LakeFront x walleye	0.154*** (0.024)		0.165*** (0.033)	-0.011	
LakeFront x smallmouth bass	-0.024 (0.028)		-0.045* (0.026)	0.021	
LakeFront x yellow perch	-0.147** (0.060)		-0.150*** (0.051)	0.003	
LakeFront x white perch	0.104*** (0.025)		0.109*** (0.026)	-0.005	
LakeFront x log(Secchi)		-0.166 (0.185)	0.283 (0.155)		-0.449**
LakeFront x Ave Temp		0.100** (0.029)	0.025 (0.030)		0.075**
LakeFront x log(TP)		0.216** (0.075)	-0.008 (0.086)		0.224**
Property and Neighborhood Characteristics	Yes	Yes	Yes		
Property Fixed Effect	Yes	Yes	Yes		
Time Trend	5-degree Polynomial	5-degree Polynomial	5-degree Polynomial		
Observations	184,744	184,744	184,744		
R2	0.89	0.89	0.89		
Within R2	0.05	0.05	0.05		

Note: This table shows the coefficient estimates of fish abundance and water quality indicators on lakefront premiums. Column (1) presents the specification with only fish abundance x lakefront dummy in the model. Column (2) presents the specification with only water quality indicators interacted lakefront dummy in the model. Column (3) presents the specification with both fish abundance x lakefront dummy and water quality indicators x lakefront dummy included in the model. Robust standard errors are clustered at the school district and year levels (two-way clustering). Standard errors are shown in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table 5: **Capitalization and Local Tax Revenue from Oneida Lake Fishery**

Panel A: Capitalization Effect of One-SD Increase in Species Abundance

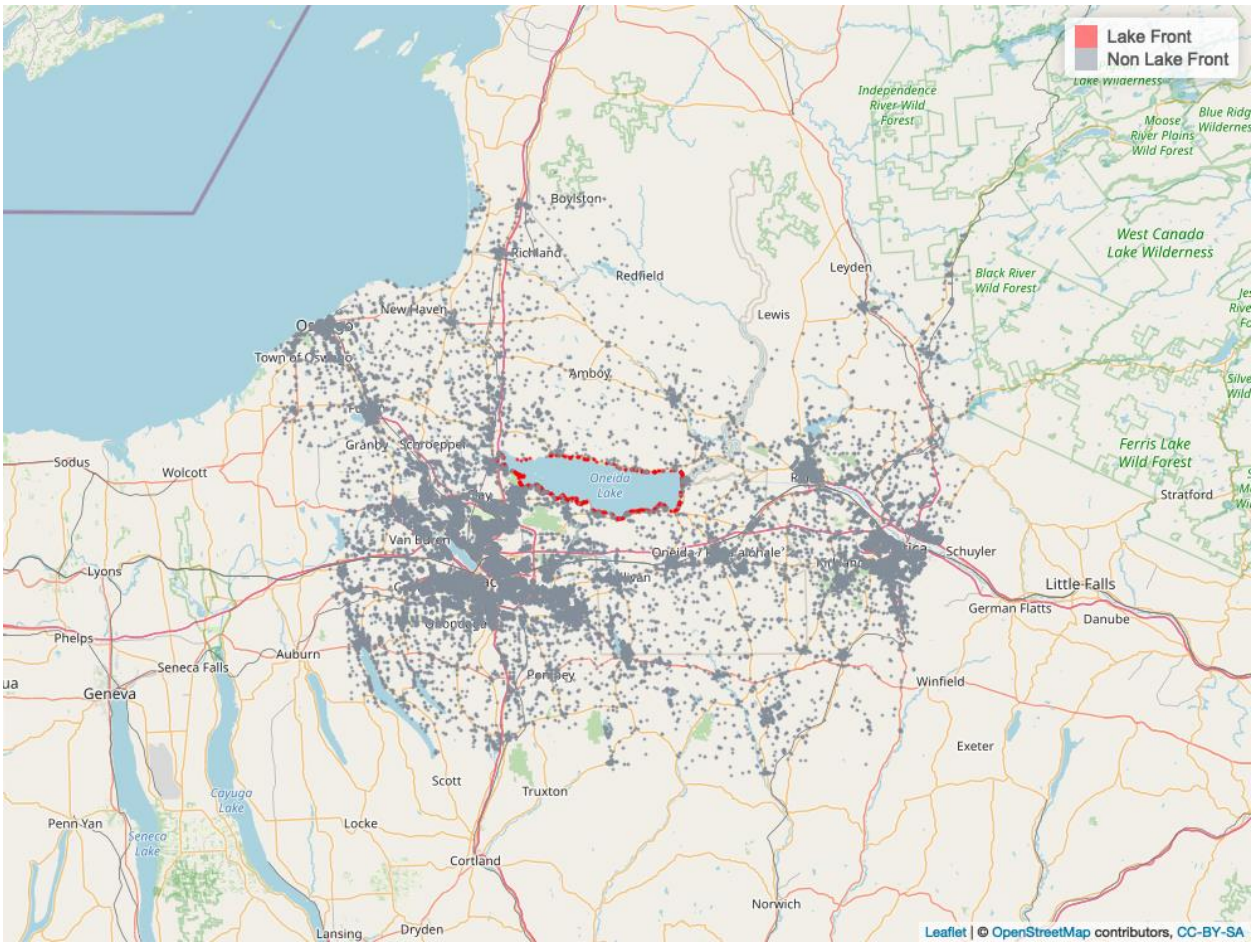
Species	Total Value of Capitalization (Million $)		Property Tax Revenue (Million $)		Annualized Rent Premium ($)	Per Trip Premium ($)
	Lake Front Property	All Property within 4km	Lake Front Property	All Property within 4km		
Smallmouth Bass	-18.47	-10.50	-0.63	-0.36	-49.4	-1.7
Walleye	64.84	70.02	2.20	2.38	872.0	29.1
White Perch	43.26	62.94	1.47	2.14	821.0	27.4
Yellow Perch	-58.90	-86.80	-2.00	-2.95	-1250.0	-41.7

Panel B: Capitalization Effect of Walleye Abundance, by County

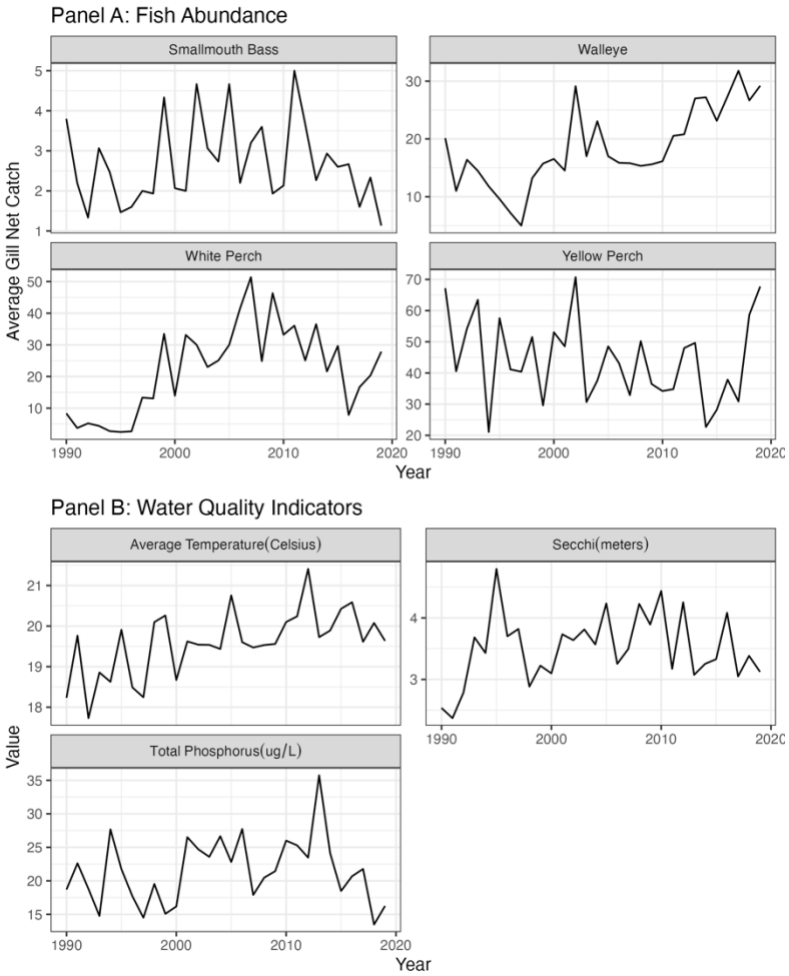
County	Oneida Lake Only			All Recreational Fishery within County		
	Number of Properties		Property Tax Revenue, +1SD Walleye (Million $)	Mean Angler Travel Distance (Miles)	State and Local Tax Revenue (Million $)	
	Lakefront	Within 4km			Angling Activity, Local Residents	Angling Activity, All Anglers
Madison	727	2,432	0.64	25.2	0.49	1.33
Oneida	454	1,620	0.43	43.7	2.79	8.44
Onondaga	638	7,134	0.82	36.1	2.28	4.47
Oswego	512	2,412	0.45	132.5	1.34	33.72

Note: This table shows total capitalization effects and local tax revenue. Panel A shows the capitalization effects from a one-standard-deviation increase of gillnet catches for four species using coefficients estimated from the spatial spillover model (Figure 2a). Property tax revenue is calculated using the average effective tax rate (i.e., levied property tax divided by assessed property value), 3.4%, for all properties within 10 km of Oneida Lake, averaged across all counties. The annualized price premium is calculated with a 3% discount rate, showing numbers only for lakefront properties. Per trip premium is calculated by assuming that the average lakefront property owner go on fishing trips 30 times per year. Panel B shows a breakdown of local economic impacts from fishery in the four counties adjacent to Oneida Lake. The property tax column shows the property tax impact of one-standard-deviation increase of walleye abundance, using effective property tax rates disaggregated by county, which range from 3.1% (Madison County) to 3.7% (Oneida County). The number of properties (regardless of transaction records) and effective tax rates are taken from assessor records. The angler travel distance and local tax revenue columns are taken from the New York Angler's Survey for the year 2017 (NYSDEC 2019). The three columns contains accounts for all recreational angling activities within these four counties, not just the Oneida Lake fishery. All property tax revenues and property premiums are discounted into 2012 dollar.

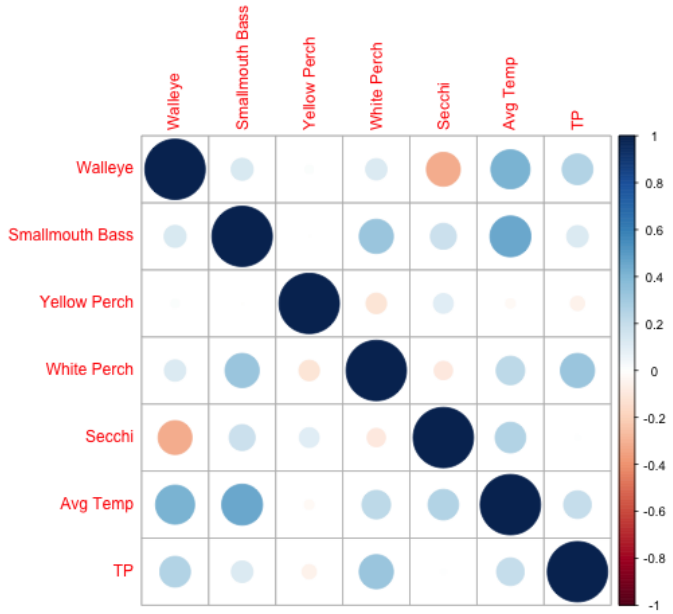
Appendix A.Additional Figures and Tables



Appendix Figure 1: Map of Repeatedly Sold Properties. Note: Each dot denotes the location of one property that has been sold at least twice during our sample period. Red dots denote properties in front of Oneida Lake (lakefront). Grey dots denote non-lakefront properties.



Appendix Figure 2: Time series of fish abundance and water quality indicators. Note: Panel A shows the average gill net catches per sample each year during summer months for four species: smallmouth bass, walleye, white perch, and yellow perch, 1990-2020. Panel B shows the annual average value of water quality indicators, averaged over summer months and across sampling locations. Units: Average Temperature: degree Celsius; Secchi: meters; TP: µg/L.



Appendix Figure 3: Correlation Matrix Between Fish Population and Water Quality Indicators. Note: Each dot denotes the correlation coefficient (between -1 and 1) between a pair of indicators. Larger-sized dots indicate stronger correlations. Red dots indicate negative correlations, and blue dots indicate positive correlations.

Appendix Table 1: Estimates with Spatially Correlated (Conley 1999) Standard Errors.

	(1)	(2)	(3)
		Fish Variables	
LakeFront x walleye	0.165*** (0.033)	0.165*** (0.026)	0.165*** (0.014)
LakeFront x smallmouth bass	-0.045* (0.026)	-0.045** (0.022)	-0.045*** (0.007)
LakeFront x yellow perch	-0.150*** (0.051)	-0.150*** (0.049)	-0.150** (0.066)
LakeFront x white perch	0.109*** (0.026)	0.109*** (0.024)	0.109*** (0.024)
		Water Quality Variables	
LakeFront x log(Secchi)	0.283* (0.155)	0.283*** (0.109)	0.283*** (0.085)
LakeFront x Avg Temp	0.025 (0.030)	0.025 (0.021)	0.025 (0.033)
LakeFront x log(TP)	-0.008 (0.086)	-0.008 (0.100)	-0.008 (0.075)
Property and Neighborhood Characteristics	Yes	Yes	Yes
Property Fixed Effect	Yes	Yes	Yes
Time Trend	5-degree Polynomial	5-degree Polynomial	5-degree Polynomial
Standard Error	School District + Year Clustered	Conley (1999) with 1 km cutoff	Conley (1999) with 5 km cutoff
Observations	184,744	184,744	184,744

Note: This table shows the coefficient estimates with different estimates for standard errors. Specifications (1) present robust standard errors in the main specification, clustered at the school district and year levels. Specifications (2) present Conley (1999)'s standard error, assuming error correlation within a 1 km distance. Specifications (3) present Conley (1999) standard errors with error correlation up to a 5 km distance. All other control variables remain the same from Table 4. Standard errors are shown in parentheses. ***p<0.01, **p<0.05, *p<0.1.